



Lesson 6.3 — The Number e

Big idea: $e \approx 2.71828$ is the base of continuous growth. It's what you get if compounding happens every instant. The function $f(x) = e^x$ is its own derivative — that's why it's the natural choice for natural processes.

Quick review

Definition: $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n \approx 2.71828\dots$

Continuous compounding: $A(t) = Pe^{rt}$ (P = principal, r = annual rate, t = years).

Continuous models: $A(t) = A_0 e^{kt}$ ($k > 0$ growth, $k < 0$ decay).

Why e shows up: the only base for which the rate of change of b^x equals b^x itself is e .

Worked example

Worked example. \$1,000 at 5% interest, compounded continuously, for 7 years.

$$A = 1000 e^{0.05 \cdot 7} = 1000 e^{0.35} \approx 1000(1.4191) \approx \$1,419.07.$$

Warm-ups

1. Estimate e^1 to two decimal places.
2. Compute e^0 .
3. Is e^x growth or decay (as x increases)?

Practice

1. \$2,000 at 6%, continuously compounded, for 5 years.
2. A bacterial population satisfies $A(t) = 500 e^{0.2t}$ (t in hours). Population after 10 hours?
3. A radioactive substance follows $A(t) = 80 e^{-0.1t}$. Amount after 5 days?

Challenge

1. How long does it take \$1,000 to double at 4% continuously compounded? (Use $\ln 2 \approx 0.693$.)
2. Compare the future value of \$5,000 over 10 years at 5% under (a) annual compounding, (b) monthly compounding, (c) continuous compounding.



Answer key — Lesson 6.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. ≈ 2.72 .
2. 1.
3. Growth.

Practice

1. $2000 e^{0.30} \approx 2000(1.3499) \approx \$2,699.72$.
2. $500 e^2 \approx 500(7.389) \approx 3,694$.
3. $80 e^{-0.5} \approx 80(0.6065) \approx 48.52$.

Challenge

1. $2000 = 1000 e^{0.04t} \Rightarrow \ln 2 = 0.04t \Rightarrow t = \ln 2 / 0.04 \approx 17.33$ years.
2. (a) $5000(1.05)^{10} \approx \$8,144.47$. (b) $5000(1 + 0.05/12)^{120} \approx \$8,235.05$. (c) $5000 e^{0.5} \approx \$8,243.61$.
Continuous gives the most.